

Dr Rachel Elizabeth Moore

Centre for Developmental Neurobiology
King's College London, Guy's Hospital Campus
London, SE1 1UL, UK

rachel.moore@kcl.ac.uk
ORCID ID: 0000-0002-5426-9545

SCHOLARLY PROFILE

My scientific interests are how neurons within the embryonic nervous system mature and develop. To address this I perform live, high-resolution time-lapse microscopy at subcellular resolution on zebrafish embryos.

EDUCATION

- 2009 – 2013 **Ph.D.**, Cell and Developmental Biology, UCL, UK
- 2008 **Honours**, Anatomy and Cell Biology, University of Melbourne, Australia
- 2004 – 2007 **Bachelor of Biomedical Science**, University of Melbourne, Australia
- 2004 – 2006 **Diploma of Modern Languages (German)**, University of Melbourne, Australia

RESEARCH EXPERIENCE

- 2022 – now **Research Associate, Houart lab**
Centre for Developmental Neurobiology, KCL, UK
- Investigating the role of SRSF RNA splicing proteins in zebrafish spinal neurons during development, including control of axonal transcripts and neuronal maturation.
 - Developing a combination of tools and live imaging techniques to observe the behavioural dynamics of RNA splicing proteins in living axons *in vivo*.
- 2014 – 2021 **Research Associate, Clarke lab**
(maternity leave Dec 2020 – Sept 2021; 3 days/week Oct 2021 – Dec 2021)
Centre for Developmental Neurobiology, KCL, UK
- Investigated neuronal polarity and axon initiation the zebrafish embryonic spinal cord *in vivo*, finding that axon initiation can occur in the absence of the microtubule cytoskeleton.
 - Identified a novel form of protrusion that mediates the spatiotemporal patterning of neuronal differentiation in the zebrafish spinal cord.
 - Helped develop the Phytochrome system for use in zebrafish, allowing precise manipulation of protein localization *in vivo* within a live, normally developing organism.
- 2009 – 2013 **Ph.D. student, Mayor lab**
Department of Cell and Developmental Biology, UCL, UK
- Identified that polarity protein Par3 affects microtubule dynamics via Rac1 signalling in *Xenopus laevis* neural crest cells, and that this is important for their migration.
- 2008 **Honours* student, Young lab** (*equivalent to a UK Masters)
Department of Anatomy and Cell Biology, University of Melbourne, Australia
- Helped determine that neuronal activity can influence neuronal differentiation during enteric nervous system development in the mouse embryo.

PUBLICATIONS

First-author research articles

Moore R.E., Pop S., Alleyne C. and Clarke J.D.W. (2022) Microtubules are not required to generate a nascent axon in embryonic spinal neurons *in vivo*. *EMBO Reports* 23:e52492

Hadjivasiliou Z.*, Moore R.E.*, McIntosh R., Galea G.L., Clarke J.D.W. and Alexandre P. (2019) Basal protrusions mediate spatiotemporal patterns of spinal neuron differentiation. *Developmental Cell* 49:907-919 (*contributed equally)

Moore R., Théveneau E., Pozzi S., Alexandre P., Richardson J., Merks A., Parsons M., Linker C. and Mayor R. (2013) Par3 controls neural crest migration by promoting microtubule catastrophe during contact inhibition of locomotion. *Development* 140:4763-75

Second-author research articles

Buckley C.E., Moore R.E., Reade A., Goldberg A.R., Weiner O.D. and Clarke J.D.W. (2016) Reversible optogenetic control of subcellular protein localisation in a live vertebrate embryo. *Developmental Cell* 36:117-126

Hao M.M., Moore R.E., Roberts R.R., Nguyen T., Furness J.B., Anderson R.B. and Young H.M. (2010) The role of neural activity in the migration and differentiation of enteric neuron precursors. *Neurogastroenterology & Motility* 22:e127-37

Review articles

Moore R.E., Clarke J. and Alexandre P. (2020) Protrusion-mediated signaling regulates patterning of the developing nervous system. *Frontiers in Cell and Developmental Biology*, 29 September 2020

Moore R. and Alexandre P. (2020) Delta-Notch signalling: The long and the short of a neuron's influence on progenitor fates. *Journal of Developmental Biology* 8:8

FUNDING & ACHIEVEMENTS

Nov 2021 **Awardee**, KCL Parents and Carers Fund (£9,668)

Nov 2019 **Awardee**, KCL Institute of Psychiatry, Psychology & Neuroscience Early Career Research Award (£2,500)

2009 – 2012 **Awardee**, UCL Graduate Research Scholarship (£56,370)

2009 – 2012 **Awardee**, UCL Overseas Research Scholarship (£39,729)

PRESENTATIONS

Invited Presentations

Feb 2024 **Seminar**, Cytoskeleton of Neurons and Glia Seminar Series, online

Feb 2022 **Seminar**, Australian Developmental Neuroscience Forum, online

March 2020 **Seminar**, Department of Anatomy and Cell Biology, University of Melbourne, Australia

Selected Abstracts

August 2024 **Talk**, International Zebrafish Conference, Japan

July 2023 **Talk**, European Zebrafish Meeting, Poland

Sept 2020 **Talk**, South-west Zebrafish Meeting, UK (2nd prize oral presentation)
Dec 2019 **Talk**, EMBO Mechanisms of Neuronal Remodelling, Israel

SUPERVISING EXPERIENCE

2023 **Supervisor** for Mehdi Talebzadeh, Wellcome Biomedical Vacation Scholar (6 weeks), KCL. I have kept in touch with Mehdi as a **mentor** as he applies to PhD programmes.

2022 **Supervisor** for Stephanie Salazar, Minority Health & Health Disparities International Research Training placement student (2 months), KCL.

2022 **Supervisor** for Claire Scott, research technician (3 months), KCL.

2019 **Daily supervisor** for Caché Alleyne, undergraduate student (9 months), KCL. Caché is a co-author on Moore R.E *et al*, (2022) and achieved 83% (a very high first-class mark) for her project.

2015 **Daily supervisor** for Ayesha Mirchandani and Ayala Shirazi, undergraduate summer students (3 months each), KCL. Ayala achieved the top mark in her class for her project.

2014 – 2015 **Daily supervisor** for Richard Taylor and Sînziana Pop, Ph.D. rotation students (3 months). Sînziana is second author on Moore R.E *et al*, (2022).

2012 **Daily supervisor** for Sara Pozzi, Ph.D. rotation student (3 months), UCL. Sara is a co-author on Moore R., *et al* (2013).

TEACHING EXPERIENCE

2022 - now **Tutor**, BSc module Developmental Neurobiology, KCL, UK. I guide a small group of students as they created a poster about “Neuronal differentiation” and present it to their peers and demonstrators.

2019 **Research Facilitator**, EMBO Practical Course: Developmental Neurobiology, London, UK. I prepared and ran a workshop demonstrating use of the Phytochrome optogenetic system in zebrafish embryos.

2017 – 2019 **Research Facilitator**, Zebrafish Development and Genetics, MBL, USA. I assisted Jon Clarke to run a workshop demonstrating use of the Phytochrome optogenetic system in zebrafish embryos.

2016 – now **Lecturer**, BSc module “Principles of Neurobiological Research” KCL, UK. I prepare and present a lecture and practical session about using zebrafish embryos in neurobiology research. The students are from a range of disciplines so the lecture is focused on comprehension and problem-solving. It was voted highlight of the module by 2022 student cohort.

2014 – 2020 **Volunteer tutor**, four A-level biology students, Access Project, UK

2014 – 2017 **Demonstrator**, MBBS module “Human Neuroanatomy”, BSc module “Developmental Biology”, BSc module “Essentials of Embryology”, KCL, UK

2009 – 2012 **Tutor**, BSc module “Principles of Genetics” UCL, UK

2009 – 2012 **Demonstrator**, BSc module “Developmental Biology”, MBBS module “Reproduction and Embryology”, UCL, UK

SCIENCE COMMUNICATION AND OUTREACH

2023 – 2024 **Curator** of CDN social media accounts

2022 – now **Committee Member**, Public Engagement Committee, Centres for Developmental Neurobiology and Neurodevelopmental Disorders KCL

2020 – now Collaboration with director and artist Suki Chan

2020 **Module leader (Lecturer)**, in2Science, UK

2018 – 2019 **Participant**, BrainWaves, Sidmouth Science Festival, UK

2016 – now **Lab Host**, DevNeuro Academy Widening Participation Project, KCL, UK

2016 **Co-Organiser**, Pint of Science, KCL's Beautiful Mind Team, UK

ACADEMIC SERVICE

2022 **Working Group Member**, Supervisor/PI Communication working group, Centre for Developmental Neurobiology KCL

2018 – 2020 **Committee Member**, Animal Welfare and Ethical Review Body (AWERB), Guy's Hospital Campus KCL

2017 – now **Peer review**, PNAS, Frontiers in Cell and Developmental Biology, Mechanisms of Development

2017 – 2020 **Organiser**, Post-doc meetings, Centre for Developmental Biology KCL

2016 – 2020 **Treasurer**, Young Embryologist Network

2011 – 2012 **Chair**, Young Embryologist Network